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Paper Code : BS-PH101/BSPH101 Physics-I
UPID : 001003

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) What is the displacement of a particle executing SHM, when the velocity is maximum?
(ii) What is poulation inversion?
(iii) What is the main difference between a polar dielectric and a non-ploar dielectric?
(iv) Phase velocity and group velocity are equal when the medium is _____
(v) MB statistics is applicable for photons. State True or False.
(vi) Define solenoidal vector.
(vii) What is Optic Axis of a crystal?
(viii) Define magnetization vector M.
(ix) What is the zero-point energy of a quantum harmonic oscillator?
(x) Which statistical distribution is followed by free electrons in a conductor?
(xi) Which property of light exhibits its transverse nature?
(xii) Calculate the energy in eV associated with a photon of Ruby laser radiation of wavelength 589nm.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. a) Plot the energy density of blackbody radiation as a function of wavelength at three different temperatures T1<T2<T3 [5]
   b) What do you mean by Ultra-Violet Catastrophe?
3. Write the differential equation for free oscillation and forced oscillation explaining each term. [5]
4. Distunguish between Amplitude resonance and velocity resonance. What is logarithmic decrement in damped oscillation? [5]
5. Find the missing orders of a double slit diffraction pattern with slit width 0.21nm and and distance between the slits is 0.63nm. [5]
6. A particle is trapped in an infinite one dimensional potential well of width 'L' . Plot its wavefunction and probability density function in its first excited state. Find the probability density at x=L/2. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Distinguish between Fresnel and Fraunhofer Diffraction [3]
   (b) Write the characteristics of LASER. [4]
   (c) Mention three methods of producing plane polarized beam of light. [3]
   (d) Explain briefly what do you mean by double refraction. [3]
   (e) State Malus law. [2]
8. (a) Define conservative force and dissipative force giving examples. Show that the force acting on a particle in simple harmonic motion is a conservative one. [6]
   (b) 2 distinguishable particles which can be in in one of the E, 2E or 3E energy states. The total energy of the system is 4E. Find all the macro states and their corresponding microstates. How many total states are possible? [5]
   (c) Indicate which statistics will be applicable for (i) electron (ii) photon (iii) pi meson (iv) $_2He^3$. [4]
9. (a) Derive the following expression [5]

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$$v_g = v_p - \lambda \frac{dv_p}{d\lambda}$$

(b) State uncertainty principle. An electron is confined in a box of width 5 nm. Find the uncertainty in momentum and velocity of the electron. (mass of the electron m=10^{-31}kg). [5]

(c) What do you mean by degeneracy? Explain with the help of 3-dimensional potential cube. [5]

10. (a) What is Grating element? Write and plot the intensity distribution of a single slit diffraction pattern and find the positions of maxima, minima and secondary maxima. [2+1+2+4]

(b) Find the minimum thickness of a half wave plate and a quarter wave plate for a light wave of wavelength 5893 Angstrom. Given $\mu_0 = 1.66$ and $\mu_e = 1.47$ [2]

(c) What are the three basic components of a LASER? What is a optical resonator? [2+2]

11. (a) Find the moment of inertia of a cylinder of mass M and radius R about its own axis. [5]

(b) Calculate the number of photons emitted per second by a 20W source of monochromatic source of wavelength 431nm. [5]

(c) Find the energy difference between the ground state and 2nd excited state of an electron in a one dimensional infinite potential well of dimension 10 angstrom($m = 9.1 \times 10^{-31}$kg). [5]

*** END OF PAPER ***

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